

WELLSENSE · REQUIREMENTS

WellSense requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wellsense		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-02	Amish Chadha	Constructable design (WLS-DDR-004); R12 reported against the \$200 value-engineering target; R2 status names the tube collar
0.5	2026-09-26	Amish Chadha	Budget approved by Amish; R12 target \$200, status Not met to Met
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from WLS-CAL-001; R1, R6 and R12 restated under WLS-DDR-001 (D1, D3, D5); assumptions updated
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

WellSense requirements

These requirements are checked by calculation in WLS-CAL-001 v0.4 against the constructable design of WLS-DDR-004 and WLS-PRC-001 v0.6. Targets remain proposals, to be revised after co-design. Under WLS-DDR-001 (decided by Amish, 2026-09-25: go with recommendation), R1 names the 0 to 10 m range (D3), R6 states how 1 min readings are sent (D5), and R12 costs the WellSense parts only, with the FieldNode core costed in the FieldNode repo (D1). Under WLS-DDR-002, R12 is restated at \$190 for the 30 m design well (raised to \$200 by Amish on 2026-09-26 to cover the priced BOM), R10 limits the probe to 22 mm in the 1 in tube, R6 adds the 30 min uplink rule for pumping tests on public networks, and R15 requires FieldNode's sun shield at hot sites. No target was relaxed. Under the 2026-10-01 portfolio rule, `budget_usd` is a value-engineering target, a hypothetical control target rather than a limit, so R12 is reported as over or under that target. On paper, 7 requirements are met, 4 are at risk, 1 is not met (R9), 3 cannot be verified at TRL 3, and R12 is \$44.40 over its value-engineering target.

Table 1. Requirements and status at TRL 3 (WLS-CAL-001 v0.4, Table 3).

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3
R1	Measurement range (water above the probe)	0 to 10 m (DDR-001, D3), with a 0 to 20 m variant	Datasheet	Met: 98.07 kPa full scale; swings to 9 m fit with 1 m headroom
R2	Installation depth below the wellhead	Probe to 60 m or more	Cable and tube length; hanger load	Met: cable factor 11.5 and tube factor 58 at 60 m; a split collar carries the tube on the seal plate (0.24 MPa bearing)
R3	Resolution	1 mm or better	ADC and shunt calculation	Met: 0.52 mm (1.04 mm on the 20 m variant)
R4	Accuracy after field calibration against a manual tape	±20 mm over the range	Calculation, then side-by-side with a manual tape	At risk: 12.5 mm root sum square, 27.1 mm worst case, with a 0.25 % class probe and two-point calibration; a 0.5 % class probe does not meet it
R5	Long-term drift	20 mm a year or less, checked by a manual reading each quarter	Field record	Not verifiable at TRL 3: drift of low-cost transducers is not published; a quarterly check leaves at most 5 mm uncorrected
R6	Reading interval	15 min default; 1 min during a pumping test, logged and sent in batches (DDR-001, D5); on a public network a test sends every 30 min, on a private Twinkit gateway every 15 min (DDR-002)	Firmware configuration	Met by design: 30 min batches use 22.7 s a day at SF9, within 30 s fair use
R7	Local storage	1 year of 15 min readings without a link	Storage calculation	Met: 1.12 MB a year, 6.7 % of FieldNode flash
R8	Energy	Probe and interface use 5 % or less of the FieldNode sensor energy budget at 15 min	Power budget	Met: 38.2 mWh a day, 1.6 % of the 100 mW allowance
R9	Water safety	All wetted parts 316 stainless steel or materials certified for drinking water contact; well sealed against surface water	Material certificates	Not met on evidence: no certificate in hand for low-cost cable, seals or probe
R10	Compatibility with the existing pump	Probe of 22 mm or less in its own 1 in access tube (DDR-002); no contact with pump, riser or pump cable	Design review, installation trial	At risk: tube and probe fit the 150 mm design well with 2.3 mm radial clearance, but fitting a tube may need the pump pulled
R11	Installation	Two trained people, hand tools, 2 h or less when an access tube already exists	Installation trial	Not verifiable at TRL 3: 120 min estimate, at the limit
R12	Cost	Value-engineering target: WellSense parts \$200 per well at a 30 m probe depth (budget_usd , a hypothetical	Priced BOM	Over the target: \$244.40 at 30 m for the constructable design (WLS-

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3
		control target), plus \$1.80 per metre beyond; FieldNode core costed in the FieldNode repo (DDR-001, D1)		DDR-004), \$44.40 over; \$383.40 with the FieldNode core
R13	Data use	Dashboard shows level below ground, daily drawdown and change against the same month last year on a basic phone; CSV export	Demonstration with sample data	Not verifiable at TRL 3 (software not started)
R14	Data ownership and privacy	Water level, time and well ID only; the hosting community controls sharing	Design review	Met by design
R15	Environment	Probe IP68 at 1.5 times range; above-ground parts IP65, -10 to 55 °C; FieldNode sun shield fitted at hot sites (DDR-002)	Datasheets, later field test	At risk: met on datasheets for WellSense parts; with its shield the FieldNode core stays at 48.5 to 52.2 °C inside at 45 °C ambient (FND-CAL-001 v0.2), but it is rated to 45 °C ambient, not 55 °C
R16	Tamper resistance	Wellhead parts lockable; no exposed cable at reachable height	Design review	At risk: the surface cable now runs in galvanized conduit and a short flexible tail from the tube cap to the junction box (DDR-002, DDR-004), but locking of the seal plate, tube cap and box is not yet specified

■ Assumptions

- The full list of assumptions is in WLS-CAL-001, Table 1. The main ones follow.
- Water density 1,000 kg/m³ nominal; 1 m of water is 9.81 kPa. Density changes by 0.29 % between 4 and 25 °C, so firmware uses a fixed site density set from the groundwater temperature at installation, and local gravity rather than standard gravity.
- A vented (gauge) transducer (DDR-001, D2), so the reading is already referenced to air pressure. The barometric sensor records the aquifer's response to air pressure and detects a blocked vent.
- Accuracy after a two-point field calibration: the tape reading at installation and again after lifting the probe a measured 1 m. Nonlinearity, hysteresis and repeatability of a 0.25 % class probe taken as 0.10 % of full scale (typical, not measured).
- Loop powered at 24 V from a boost on the interface board, fed by FieldNode's switched 12 V rail at 90 % efficiency; 22 mA for 2 s per reading.
- Design case: 150 mm casing, 4 in pump on a 1-1/4 in riser, probe 30 m below the top of the casing.